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Performance Analysis of Diversity Selection Combining on Radio Frequency for Image Transmission

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Abstract. This paper describes a simulation of the image transmission system with diversity selection combining (DSC) method on radio frequency. DSC is used to obtain multiple data streams corresponding to the transmitted image at the receiver in order to combat errors during image transmission on wireless channels. These individual image data streams are combined to form a composite image with higher perceptual quality. When DSC is used, it is expected that it can affect the number of errors, so that the signal transmitted to the wireless channels becomes more robust against interference caused by transmission channels. By using DSC technique, it can increase the quality of image transmission at the receiver. The result of the experiment which was carried out to test bit error rate (BER) and peak signal to noise ratio (PSNR) on Lena image with rate 0.8 bpp by using DSC on radio frequency in Rayleigh fading channel showed that there was a decrease in BER at the average of 0.0152 or 3.39%, while for PSNR there was an increase at the average of 1.6737 dB or 6.73% when comparing to the system without DSC.

1. Introduction

The advancement of digital image technology has triggered a rapid development in the imaging business, which leads to high demand for effective digital image storage and transmission media [1]. It happens because digital images is a type of information that has a relatively larger size than other types of information, such as text and sound [2]. In this case, compression or coding of sources becomes a solution to reduce the number of information symbols, so that, it can streamline storage media and increase the transmission rate for certain channel capacities [3]. On the other hand, with a certain transmission rate, compression allows the use of channels with a smaller bandwidth.

Set Partitioning in Hierarchical Trees (SPIHT) wavelet-based image compression method [4] is the development of the embedded zerotree wavelet (EZW) method [5]. SPIHT has better performance and simpler algorithm than EZW, which using partitioned grouping for significance tests in accordance with the decreasing threshold order. With the reduced number of symbols to represent an image, the compressed image will experience the significant impact on information damage due to noise than uncompressed images. Therefore, coding controls and error correction is absolutely necessary for transmission of compressed images [6-10]. Reed Solomon (RS) codes is one of the alternative codings that has comparatively capabilities on correcting additional redundant data. This encoding is based on bit groups [11-12], so that it has ability in error correction, especially burst errors. In wireless



communication, the signal can propagate from the transmitter to the receiver through various reflection paths, which is called multipath propagation [13-16]. This phenomenon causes fluctuations in signal quality in the receiver which is known as fading. Fading is a noise that contaminates the signal by multiplying the signal, thus it gives a greater impact than additive noise. Diversity technique is one of the techniques that is often used to reduce signal quality degradation due to this phenomenon. Diversity is used to improve the performance of radio channels without increasing transmission power [17].

Several techniques have been used to improve the performance of systems such as the use of diversity technique equal gain combining on radio frequency [15-17]. However, it is not a good technique because the increment of signal to noise ratio (SNR) will not automatically increase PSNR, but it only implies to degrade BER [18]. The use of diversity maximum ratio combining on wavelet domain improves the performance of image reconstruction [18]. Diversity techniques use on wavelet domain is to get better reconstruction image that used a two-state Gilbert-Elliott channel [18].

For this research purpose, DSC on radio frequency [2] uses fading distribution reyleigh and noise which has normal distribution gaussian [2]. To combat the errors that occurs on wireless channel, the use of coding channel is advantageous. In this research, RS codes and DSC are chosen due to its high ability in detecting and correcting errors on the image that caused by fading and noise.

The experiment is done by comparing the system with and without using DSC. The evaluation of the improvement quality system on the image transmission is calculated by the use of BER and PSNR. BER indicates how many bits with errors or symbols occurred at the receiver and compares to the number of bits or symbols received in a transmission, while PSNR is used to evaluate the quality of reconstruct image [4-7,18].

In the simulation, image information with size 8 bpp is compressed by SPIHT with bit rate 0.8 bpp. The purpose of using SPIHT compression is to minimize signal bandwidth so that the use of transmission channel become more efficient. Diversity selection combining technique is used to improve image quality on the receiver because this technique has the ability to choose the largest SNR. RS codes (31.15) [18-19] is used to protect data during transmission process.

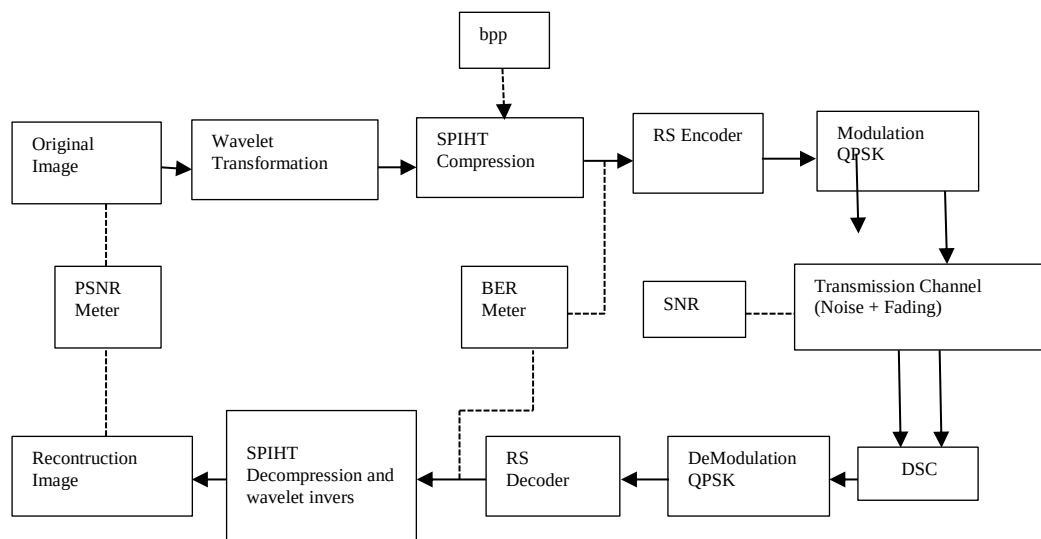


Figure 1. The design system of SPIHT compressed image transmission with DSC technique in Radio Frequency Domain.

The design system of SPIHT compressed image transmission with DSC technique on Radio Frequency Domain can be seen in figure 1. The image to be transmitted is a grayscale image of 512×512 with a depth of 8 bits. Before compressing the image, it is decomposed by wavelet transformation [18]. The wavelet transformation image is compressed with the SPIHT algorithm with bit rate 0.8 bpp [4], so that

it produces the series of bit streams with the number corresponding to the bits per pixel (bpp) which is independent variable of the system. To reduce errors that occur during transmission, the RS coding is used. In this system, the output of RS Codes is then modulated QPSK before it is transmitted to the wireless channel.

The process of DSC uses two diversity channels by implementing error correction and grouping the bit stream into blocks can be seen on Figure 2. The algorithm of this diversity is based on the block chosen of bit $b(l)$ from one bit stream which based on the characteristic of wavelet transformation. The algorithm of DSC captures two signals that are independent on fading and noise from the transmission channel. Then, this DSC choose signal that has largest SNR. The simulation model for the simulation system is as follows [2]:

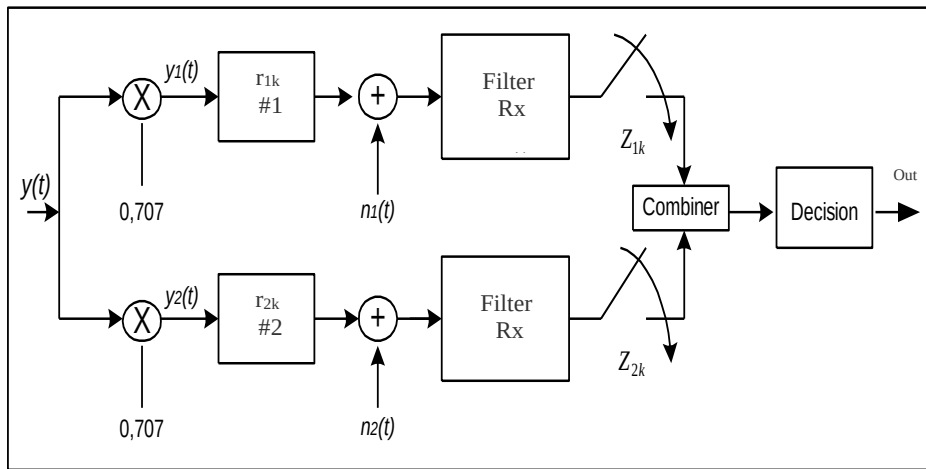


Figure 2. The model system with Selection Diversity Combining (DSC) [2].

On decision circuit, decoding process is used to detect bit on the receiver. This process is done by applying a rule. The rule gives the optimum performance, so it could reduce the error [2]. The simulation model for DSC is as follows:

$$Z_k = \begin{cases} Z_{1k}, & \text{if } |Z_{1k}| \geq |Z_{2k}| \\ Z_{2k}, & \text{if } |Z_{2k}| \geq |Z_{1k}| \end{cases} \quad (1)$$

where :

Z_k = detected Bit on receiver

Z_{1k} and Z_{2k} are decision variables at the first and second diversity paths and Z_k is the decision variable at the output of the diversity combiner.

The diagram block for this method is shown on figure 2. If there are some diversity branches for signal that gets into chosen circuit, so the output of this circuit Z_k is the chosen of the largest SNR. On fading channel, some diversity branches of Rayleigh channel at the receiver and assumed on every branch has the same average value of SNR, it can be formulated as follows [2]:

$$SNR = \Gamma = \frac{E_b}{N_0} \overline{\alpha^2} \quad (2)$$

where $\overline{\alpha^2}$ is mean from the random variabel. In this case, the random variable is the envelope from fading Rayleigh. If each branches has instant SNR γ_i , the power of spectral density (PDF) from γ_i is [2]:

$$p(\gamma_i) = \frac{1}{\Gamma} e^{-\frac{\gamma_i}{\Gamma}} \quad \gamma_i \geq 0 \quad (3)$$

where Γ is average SNR from every branches. Probability that a branch has SNR less than the value of threshold γ is [2]:

$$P_r[\gamma_i \leq \gamma] = \int_0^\infty p(\gamma_i) d\gamma_i = \int_0^\infty \frac{1}{\Gamma} e^{-\frac{\gamma_i}{\Gamma}} d\gamma_i \quad (4)$$

Probability that every branches recieved simultaneous signal less than SNR threshod γ is [2]:

$$P_r[\gamma_1, \dots, \gamma_M \leq \gamma] = (1 - e^{-\frac{\gamma}{\Gamma}})^M = P_M(\gamma) \quad (5)$$

$P_M(\gamma)$ is probability for all branches that fail to get the value of SNR = γ . If the only branch reached SNR > γ , the probability become [2]:

$$P_r[\gamma_i > \gamma] = 1 - P_M(\gamma) = 1 - (1 - e^{-\frac{\gamma}{\Gamma}})^M \quad (6)$$

From the equation above, for the probability of SNR more than threshold γ when selection diversity is used. To indentify average of signal to noise ratio from received signal when using diversity, it needs pdf from signal fading. For selection diversity, the average SNR is gained from the first $P_M(\gamma)$ [2].

$$P_M(\gamma) = \frac{d}{d\gamma} P_M(\gamma) = \frac{M}{\Gamma} (1 - e^{-\frac{\gamma}{\Gamma}})^{M-1} e^{-\frac{\gamma}{\Gamma}} \quad (7)$$

So, the average SNR ($\bar{\gamma}$) is formulated as follows [2]:

$$\bar{\gamma} = \int_0^\infty \gamma P_M(\gamma) d\gamma = \Gamma \int_0^\infty Mx (1 - e^{-x})^{M-1} e^{-x} dx \quad (8)$$

where x is the comparison between γ and Γ .

The above equations are used to evaluate the increase of the average SNR given by selection diversity. DSC offers the improvement on link margin without needing the added power on transmitter. This methods is easy to be implemented, but it did not result the optimum result since it did not use all possible branches.

2. Materials and methods

This research is done by doing a simulation method with the Matlab program to get the optimum performance and system reliability, as well as design accuracy before it is implemented. The transmission method used in this research is RS Code (31, 15) in the coder and the decoder [18-19]. The diversity technique used is DSC [2] on the Radio Frequency and the channel model used is additive white Gaussian noise (AWGN) and Rayleigh Fading [2].

2.1. System performance test

The system performance test in this study uses two parameters, namely BER and PSNR. BER is used to measure how many errors occurred at the receiver. PSNR is used to determine the quality of the reconstructed image at the receiver. BER and PSNR parameters can be explained as follows.

2.1.1. The Calculation of Bit Error Rate (BER)

The bit error ratio (BER) is measured by dividing the number of bit errors by the total number of transferred bits. The measurement of BER can be given by the following equation [2]:

$$BER = \frac{n}{N} \quad (9)$$

where: n = number of incorrect bits of information at the receiver
 N = number of bits of information transmitted.

2.1.2. Peak-to-peak Signal to Noise Ratio (PSNR).

Peak Signal to Noise Ratio (PSNR) is a useful parameter to determine the quality of reconstruction image. PSNR calculation is done by comparing the pixel value of the original image with the pixel value

of the reconstructed image. PSNR shows the quality of the reconstructed image. The higher the PSNR value obtained, the better the quality of the image. The PSNR can be formulated as follows [4, 18].

$$PSNR = 10 \log_{10} \frac{255^2}{\frac{1}{N} \sum_i \sum_j (p(i, j) - \hat{p}(i, j))^2} \quad (10)$$

where: $p(i, j)$ is the original image pixel value,

$\hat{p}(i, j)$ is Pixel value of reconstructed image,

N is the number of pixels in image, while the 255 value is the highest value for grey scale image.

2.2. Research Variable

The variables used in this study are as follows:

1. Independent variables are the variables that can be manipulated. The independent variables used in the study are:
 - a. Bit per pixel (bpp) where this variable represents the average number of bits for each pixel of the compressed image.
 - b. SNR where this variable is a comparison of signal power and noise power in the transmission channels.
2. Dependent variables are the variable that changes based on the changes in independent variables. The dependent variables used in the study are:
 - a. BER where this variable is a comparison between the numbers of bits error with the total number of bits received. This comparison is done between the bit stream SPIHT compression results in the transceiver with the bit stream of the decoding result of Reed Solomon in the receiver.
 - b. PSNR where this variable represents the degree of integrity of the reconstruction image

3. Results and discussion

To evaluate the system performance analysis of DSC on radio frequency in Rayleigh fading channel, several experiments were carried out to test the BER and PSNR value results for Lena image with rate 0.8 bpp. BER test results are given in table 1 and Figure 3 and Figure 4.

Table 1. The performance analysis of DSC on radio frequency in Rayleigh fading channel with rate 0.8 bpp.

SNR (dB)	Without DSC		With DSC	
	BER	PSNR (dB)	BER	PSNR (dB)
10	0.0216	14.6946	0.0204	14.7464
12	0.0175	19.3743	0.0170	19.7135
14	0.0153	21.2786	0.0148	24.5918
16	0.0141	24.6029	0.0139	26.2011
18	0.0132	27.1660	0.0129	30.2538
20	0.0126	32.0900	0.0123	33.7379

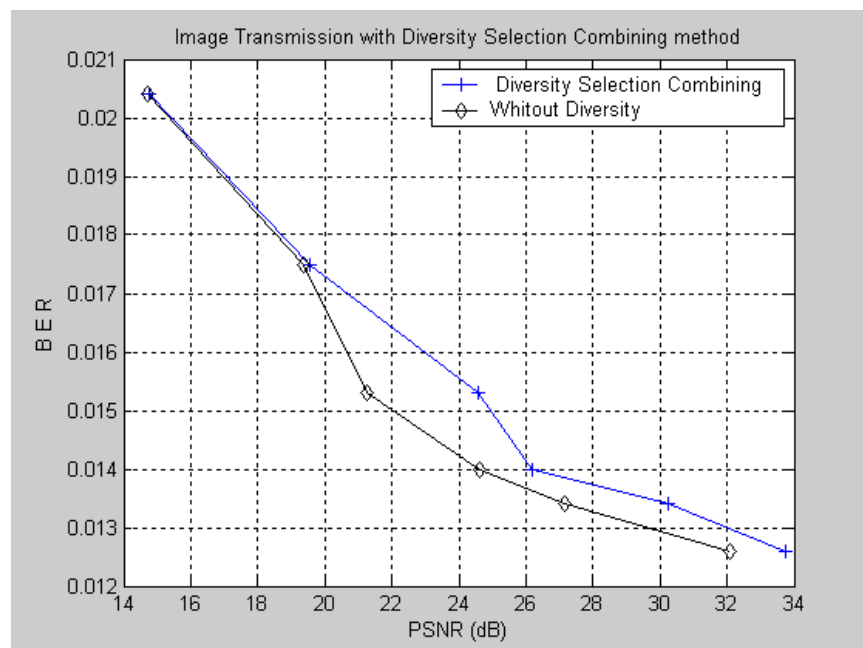
Table 1 represents the comparison of the DSC performance on radio frequency in Rayleigh fading channel with rate 0.8 bpp, with SNR 10dB, 12dB, 14dB, 16dB, 18 dB and 20dB with and without DSC. From the table, the system with DSC shows the improvement on the system performance. It can

be seen on the error that decreased to the average of 0.0152 or 3.39 %, while for the quality of image reconstruction is increased to the average of 1.6737 dB or 6.73%.

Without Diversity SNR 14 dB	Diversity selection combining SNR 14 dB
	
BER = 0.0153 PSNR = 21.2786 dB	BER = 0,0148 PSNR = 24.5918 dB

Figure 3. The comparison result of image reconstruction with and without DSC.

Figure 3 shows the comparison of BER and PSNR with and without DSC. The image with SNR 14 dB is the representation of one of the image reconstructions. For the system with DSC, there is a significant improvement at 14dB SNR in image reconstruction quality compared to other SNRs. This significant improvement is seen in the decrease of error by 0.0005 or 3.4%, and an increase in the quality of image reconstruction by 3.3132 or 13.47%.



Graph 1. The comparison of graphs of the system performance with and without DSC.

Graph 1 represents the comparison of the DSC performance on radio frequency in Rayleigh fading channel with rate 0.8 bpp and with SNR 10dB, 12dB, 14dB, 16dB, 18 dB and 20dB with and without DSC. Comparing to system without DSC, the system with DSC mengalami perbaikan sistem yang dapat

dilihat dari penurunan error, sedangkan kualitas image reconstruction terjadi peningkatan yang signifikan kecuali pada SNR 10 dB dan 12 dB. Comparing to the system without DSC, the system with DSC has an improvement in the system performance which could be seen from the decrease in error, while the quality of image reconstruction is improve significantly except for SNR 10 dB and 12 dB.

4. Conclusion

The result of the experiments which were carried out to test BER and PSNR on Lena image with rate 0.8 bpp by using DSC on radio frequency in Rayleigh fading channel showed that there was a decrease in BER at the average of 0.0152 or 3.39%, while for PSNR there was an increase at the average of 1.6737 dB or 6.73%. However, the SNR does not always make the PSNR increase significantly, this can be seen from SNR 10 dB and 12 dB. However, BER decreases in every SNR tested. The lowest decreasing of BER was at SNR 10 dB. It happened because the SNR is able to combat the disturbance of fading and noise. However, at SNR 10dB and 12 dB, the slightly improvement was caused by the random nature of the fading and noise which was very dominant to the information signal. While at SNR 20 dB, the improvement for both BER and PSNR were not significant because they had reached a maximum value. The results of this study show that the use of DSC in the Radio Frequency can reduce errors and improve the quality of image reconstruction. In order to make this research be more optimal, it is recommended to develop several systems such as modulation technique improvements, error control coding techniques, and standard source coding techniques.

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